Forecast-aware route planning for military vehicles on the Pachmarhi plateau

A from-scratch explainer of the project, the data, the decisions made so far, and the plan for the ICDCN 2027 poster submission.

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Project folder	Documents\meteorology

How to read this

Each section builds on the one before, so it reads best front to back. Blue boxes explain an idea in plain words. Orange boxes are warnings or things that could go wrong. Green boxes are decisions already made. Purple boxes are formulas, for when you want the exact version; you can skip them and still follow everything. There is a glossary at the end.

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1. The problem in one page

Military vehicles often have to leave the road: to get around a blocked bridge, to reach a position with no road to it, or because the road is too exposed. Off the road, the ground itself decides whether a vehicle can move. Dry, firm soil carries a heavy truck easily. The same soil after three days of monsoon rain can turn to mud that swallows the wheels, and the vehicle is stuck.

Whether a piece of ground can carry a particular vehicle is called **trafficability**. Planners describe it with three labels:

Label	Meaning
GO	The vehicle crosses normally, and many vehicles can follow in its tracks.
SLOW-GO	It gets across, but slowly, leaving ruts. Only a few vehicles can use the same track.
NO-GO	The vehicle gets stuck (or cannot climb, or cannot cross water).

Two things make this hard:

- **It changes over time.** The ground is firm in May, soft in July, and firm again by November. So a map of 'where can trucks go' is only correct for one day.
- **It depends on the vehicle.** A light jeep can cross ground that would trap a loaded 18-tonne truck.

> **The question this project answers**
>
> **A vehicle (or a convoy) has to cross the Pachmarhi plateau over the next few days. Which route avoids getting stuck, given that the ground will be wetter or drier by the time the vehicle gets there?**

Most earlier work plans a route using the conditions *today*. Our idea is to plan using a *forecast* of the conditions on the days the vehicle will actually be travelling, and then to prove, using past years, that this avoids routes that looked fine on departure day but failed later.

Pachmarhi is a hill station on the Satpura plateau in Madhya Pradesh, about 1,067 m above sea level. It has a strong monsoon (most rain falls June-September), so the ground there swings from very firm to very soft every year. That makes it a good test site.

2. Ideas you need first

This section explains every idea the rest of the document uses. Nothing here is specific to our data yet.

2.1 Measuring how firm the ground is: the cone index

Soldiers and engineers measure ground firmness with a **cone penetrometer**: a metal rod with a small cone on the tip. You push it into the soil and read how hard you had to push. Hard pushing means firm ground. The reading is called the **cone index**.

The **Rating Cone Index (RCI)** is a version of that reading that also accounts for how much the soil weakens when vehicles knead it. Higher RCI means firmer ground. In this project, RCI is the single number that describes 'how firm is the ground today'.

2.2 Measuring what a vehicle needs: the vehicle cone index

Every vehicle has a **Vehicle Cone Index (VCI)**: the minimum RCI it needs to get through. Heavy vehicles with small, narrow tyres need firm ground (high VCI). Light vehicles with wide tyres or tracks can manage softer ground (low VCI). There are two versions:

- **VCI1**: the firmness needed for *one* vehicle to pass once.
- **VCI50**: the firmness needed for *fifty* vehicles to pass along the same track. This is higher, because each vehicle churns the soil and leaves it weaker for the next.

The rule that turns firmness into GO / SLOW-GO / NO-GO

Compare today's ground (RCI) with the vehicle's needs (VCI):

- RCI at least VCI50: **GO** (even a long convoy is fine)
- RCI between VCI1 and VCI50: **SLOW-GO** (a few vehicles, ruts)
- RCI below VCI1: **NO-GO** (the first vehicle gets stuck)

Watch out: units

Cone index numbers are traditionally written in pounds per square inch (psi). Our code labels them as kilopascals (kPa), which is a different unit (1 psi is about 6.9 kPa). The numbers in the code (VCI 20-60, dry ground about 300) match the psi convention. In kPa, 40 would mean almost liquid mud. A reviewer who knows soil mechanics would notice, so before writing the abstract we relabel everything as 'cone-index units'.

2.3 Terrain: slope, roughness, land cover

- **Slope**: how steep the ground is, in degrees. Every vehicle has a maximum slope it can climb.
- **Roughness** (measured by the Terrain Ruggedness Index, TRI): how bumpy the ground is. Rough ground slows vehicles down.
- **Land cover**: what is on the surface: forest, water, bare soil, crops, buildings, roads. A wheeled vehicle cannot drive through dense forest or a lake.

2.4 Forecasting: a few basic terms

- **Forecast horizon**: how far ahead we predict. 'h=7' means seven days ahead.
- **Persistence**: the laziest possible forecast, which says 'tomorrow will be the same as today'. Any real model must beat it, otherwise it is useless.
- **RMSE** (root mean squared error): the typical size of a forecast's mistake. Smaller is better.
- **Skill**: how much better than persistence a model is, in percent. +24% means the model's errors are 24% smaller than persistence's.
- **Training / validation / test**: we teach the model on old years (training), tune it on middle years (validation), and grade it on recent years it has never seen (test). Grading on unseen years is what makes a result trustworthy.

2.5 Judging a GO / NO-GO call: recall and precision

Imagine a model that warns 'NO-GO' on some days. Two questions matter:

- **Recall**: of all the days that really were NO-GO, how many did it warn about? Low recall means vehicles get sent out and get stuck. This is the dangerous mistake.
- **Precision**: of all the days it warned about, how many really were NO-GO? Low precision means vehicles stay home on days they could have moved. This is the costly mistake.

You can always get perfect recall by shouting 'NO-GO' every day, but then precision collapses and nothing ever moves. So **both numbers must always be reported together**. This rule matters later in the document.

2.6 Route planning on a grid: A*

We cut the map into a grid of small square **cells** (for example 30 m by 30 m). Each cell gets a **cost**: cheap if the ground is easy, expensive if it is slow, and infinite (forbidden) if it is NO-GO. A route is a chain of neighbouring cells from start to finish, and its cost is the sum of its cells' costs.

A* (pronounced 'A-star') is a classic algorithm that finds the cheapest route without trying every possible path. It always explores the most promising cells first, guided by a guess of the remaining distance to the goal. It is fast, exact, and used everywhere from video games to robots.

2.7 Fuzzy sets

In ordinary ('crisp') logic, something is either in a group or not: a slope is either 'steep' or 'not steep'. **Fuzzy sets** allow degrees: a 12° slope can be 'steep' to degree 0.4. That degree, between 0 and 1, is called its **membership**.

Fuzzy logic also has its own versions of AND and OR. The most common fuzzy AND is just the **minimum**: 'passable by slope AND passable by soil' is as true as the weaker of the two. Section 7.5 explains why this suits our problem.

2.8 Edge computing

Edge computing means running software on a device out in the field (on a vehicle, a drone, or a small computer at a forward base) instead of in a distant data centre. It matters when the network is slow, unreliable, or jammed, which is normal in military operations. A model that is small and needs no internet connection can run at the edge.

3. The conference: what ICDCN wants

ICDCN is the International Conference on Distributed Computing and Networking. The 2027 edition (the 28th) is at NITK Surathkal, 4-7 January 2027. Besides full papers it has a **poster track** for early-stage work.

Item	Detail
What you submit	A **2-page extended abstract**, two-column ACM format, 9-pt font
Review	Single-blind (reviewers see author names; put names and affiliations on page 1)
Where	EasyChair, track 'ICDCN 2027 — Poster'
Deadline	**15 October 2026**. Notification 7 Nov. Camera-ready and registration 14 Nov.
Judged on	"technical merit and innovation as well as the potential to stimulate interesting discussions"
Results needed?	"Preliminary results are sufficient. Completed work is not required."
Presentation	**In person only**, on an easel. No remote presentation.
Publication	ACM Digital Library, as a 2-page extended abstract. ACM may charge an article fee.

3.1 The catch: this is a computing-and-networking conference

ICDCN's two tracks are **Distributed Computing** and **Networking**. A poster that is only 'we forecast soil strength with machine learning' is outside its main subject. But applied work does get accepted when it is tied to the conference's topics. The call for papers lists, among others: *machine learning applications, edge computing, cyber-physical systems, vehicular networks, UAV networks, sustainability applications*.

Evidence from past years that applied work like ours fits:

Year	Accepted work that resembles ours
2027 main track	Edge-AI framework for insect pest monitoring; IoT framework for forecasting calf disease; deep-RL robot navigation without GPS; traffic-signal control
2026 posters	Short-term crowd prediction; neural surrogate for autonomous-driving communication; LLM-based driving behaviour planning; LoRa emergency network scheduling
2025 posters	About six posters and demos in total, so the track is small

Decision: how we frame the work

We present the planner as a system that runs **on a vehicle or forward base with no internet and no weather service**, using only the local weather history. Our forecasting model was already built under exactly that rule. We back it with systems numbers: model size, run time on an ordinary processor, and how few bytes are needed to share a forecast with a convoy. The convoy experiment (Section 6, Step 6) adds a genuinely distributed problem: several vehicles planning on a shared map that they change as they drive.

4. What we have: the data, file by file

4.1 The weather record: pachmarhi_daily_2015_2025.csv

One row per day from 1 January 2015 to 31 December 2025 (4,018 days, no missing values). The columns include rainfall, temperatures, humidity, wind, pressure, cloud cover, and, most importantly, **soil moisture** in two layers (0-7 cm and 7-28 cm deep).

- Rainfall for 2015-2024 comes from the India Meteorological Department (IMD), built from rain gauges. 2025 comes from ERA5, a global weather reconstruction, because IMD's archive lags about a year.
- Temperature, humidity and soil moisture come from **ERA5-Land**, a computer model of the land surface that is steered by real observations. Its grid cell is about 9 km across and sits at 1,064 m, very close to Pachmarhi's real height.
- Important: ERA5-Land soil moisture is a *model output*, not a measurement. Everything built on it inherits that.

4.2 From weather to ground firmness (already built)

The weather file has no trafficability column, so the project derives one. The chain is:

rain and evaporation → **soil moisture** (two layers, already in the file) → **soil wetness index** (the two layers combined, 35% top / 65% deeper) → **RCI** (wetter soil, lower firmness) → compared with each vehicle's **VCI** → **GO / SLOW-GO / NO-GO**

The exact formula (optional)

RCI = 300 × exp(−2.485 × SWI) × rain_factor, where SWI is the soil wetness index (0 = bone dry, 1 = saturated) and rain_factor briefly lowers firmness after recent rain.

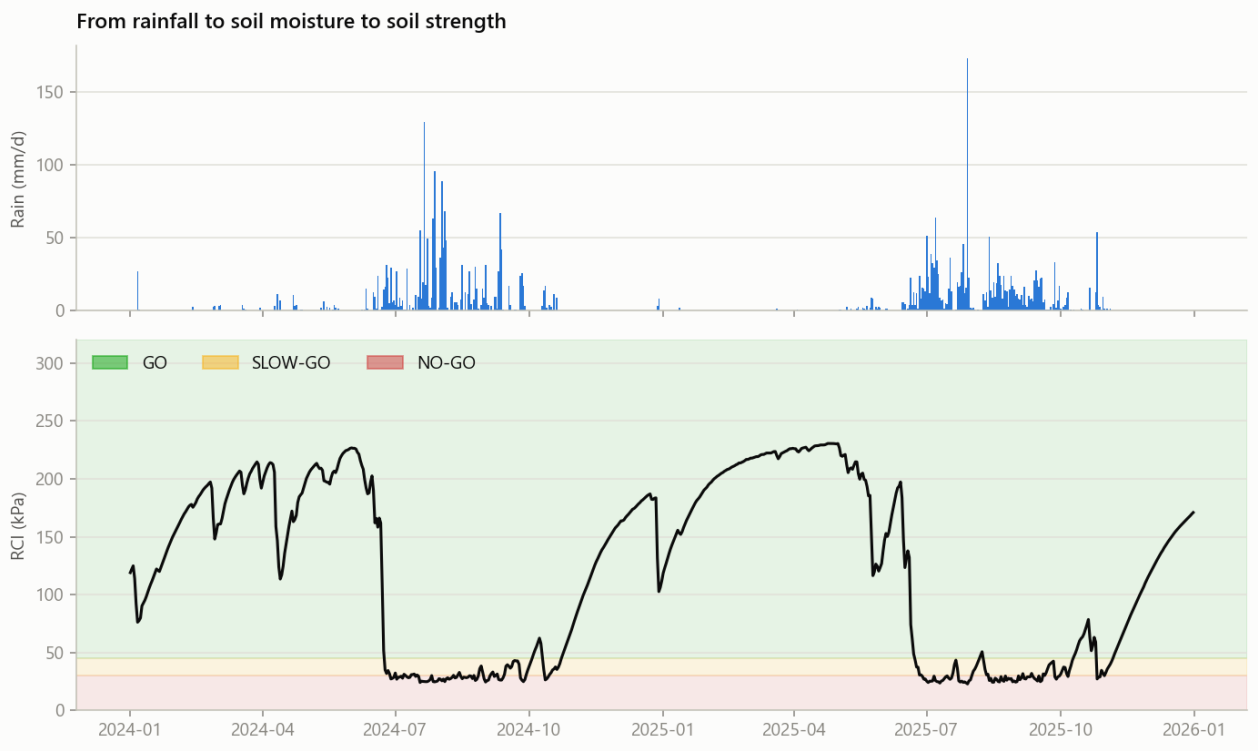


Figure 1. Rainfall (top) and the resulting ground firmness, RCI (bottom), for 2024-2025. Each monsoon drives the ground from firm (green) down into SLOW-GO and NO-GO (orange, red) within days. The axis label says kPa; as Section 2.2 explains, these are really cone-index units.

4.3 The satellite maps: maps.zip (from Google Earth Engine)

Google Earth Engine (GEE) is Google's free platform for satellite data. The zip contains 15 map layers of the Pachmarhi area, each for 3 seasons (monsoon, summer, winter): 45 images in total.

Layer	Source	What it tells us
dem_slope_deg	Elevation model	Steepness in degrees
dem_aspect_deg	Elevation model	Which direction the slope faces
dem_tri	Elevation model	Roughness (Terrain Ruggedness Index)
lst_c	Thermal satellite	Land surface temperature, °C
s1_vv, s1_vh, s1_vv_vh_ratio	Sentinel-1 radar	Radar brightness. Responds to surface wetness and roughness, and works through cloud
s1_vv_glcm_* (4 layers)	Sentinel-1 radar	Texture measures (how patchy or uniform the radar image is)
s2_ndvi	Sentinel-2 optical	Vegetation: high = dense green plants or forest
s2_ndwi	Sentinel-2 optical	Water: high = open water
s2_bsi	Sentinel-2 optical	Bare soil
s2_ndsi	Sentinel-2 optical	Snow index (little use at Pachmarhi)

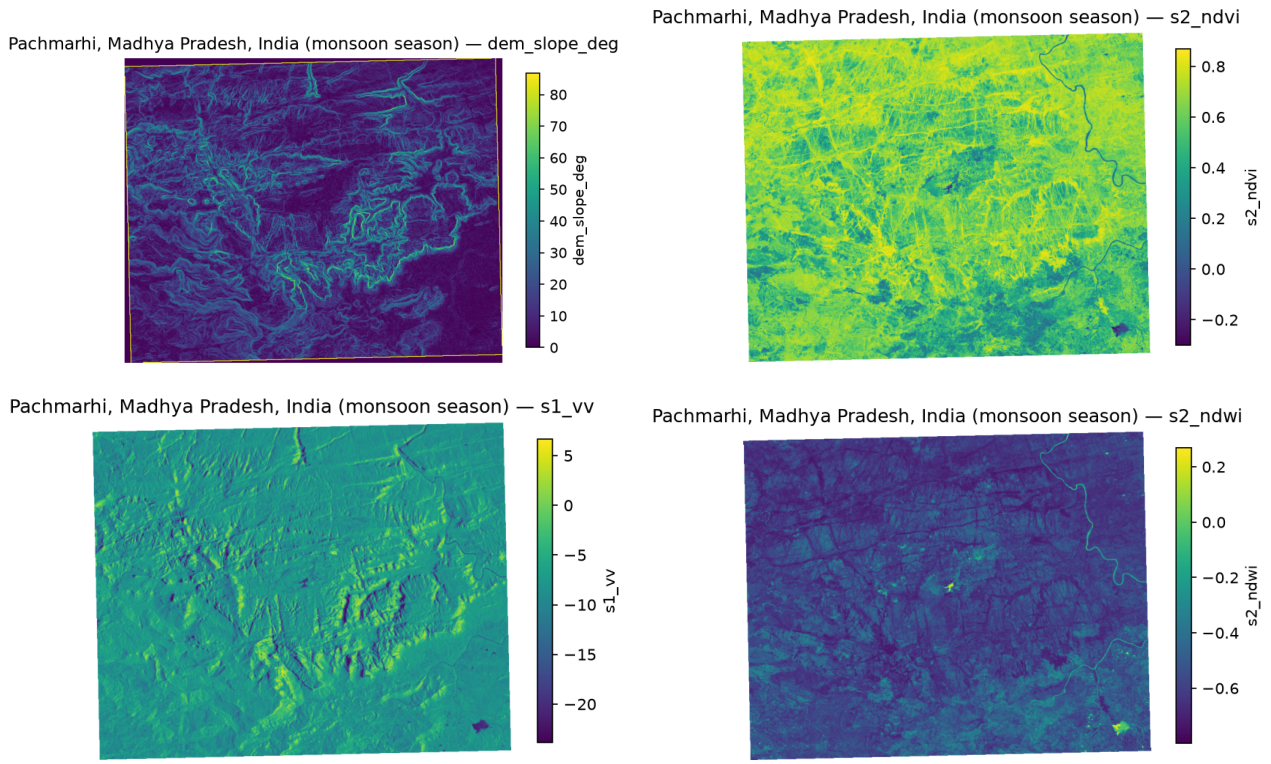


Figure 2. Four of the layers, monsoon season: slope (top left; bright lines are cliffs), vegetation (top right), radar brightness (bottom left), water (bottom right; the river on the right edge).

Are the images labelled?

No, and they don't need to be. These are not photos to sort into categories. Each one is a *measurement layer*: a map of one quantity. We use them as inputs, the same way a weather file is an input.

Watch out: these are pictures of the data, not the data

Each file is a rendered chart, with a title, a colour scale and a colour bar, rather than the underlying numbers. Colours can be turned back into numbers only roughly, and the images carry no location information. Because the layers came from GEE, the fix is easy: **re-export them from GEE as GeoTIFF files**, which store the actual values and their map coordinates. You have agreed to do this.

4.4 The vehicle file: army_vehicles_attributes.csv

2,451 rows, one per vehicle, with 34 columns such as mass, length, engine power, top speed, armour and weapons.

What this file actually is

The data comes from the video game **War Thunder** (it is a 'datamine' of the game's files). Evidence: game-engine fields such as bone_gun and fxEvent, and joke entries such as 'santa', 'space' and a 300 kg 'fake Sherman'. Other problems:

- Columns are shifted in many rows: a Kübelwagen jeep shows as 60,000 kg, with 'truck' in the horsepower column.
- About 160 vehicles have a placeholder mass of exactly 5,500 kg (a Boxer APC really weighs about 33 t).
- Top speed is a placeholder 60 or 75 km/h for most rows.
- Length, width, ground clearance and crew are 100% empty. There are no tyre sizes.
- There are no Indian vehicles and no column saying what class a vehicle is.

Decision: use it anyway, carefully

This is preliminary work and the vehicles do not need to be Indian, so we use the file. We keep only the columns that are sensible (mostly mass and engine power), drop the broken rows, and name the source in one line of the abstract so reviewers are not surprised.

Which vehicles. We originally said 'Class B', which in Indian Army terms means unarmoured support vehicles (trucks and jeeps). That was based on the assumption that the file contained only those. The actual goal is **vehicles that can stay on roads**, which we read as **all wheeled vehicles**, armoured or not: trucks, jeeps, HMMWVs, and wheeled armoured carriers such as the BTR-80, LAV and Boxer. Tracked vehicles (tanks) are left out. Examples of clean rows found so far:

Vehicle	Mass (from the file)	Engine power
ZIS-30 truck	5.8 t	50 hp
ZIS-12 truck	7.0 t	70 hp
Dodge M37	8.75 t	150 hp
ZIS-43	8.75 t	90 hp
GMC CCKW truck	11.05 t	94 hp
M939 truck	11.5 t	missing
KamAZ-4310 truck	15 t	missing
M978 fuel tanker	18.3 t	missing

Wheeled armoured vehicles are still to be filtered; many of them carry the 5,500 kg placeholder and will be dropped or corrected from public specifications.

4.5 The inspiration paper: Liu et al. (2025)

Construction of dynamic trafficability map for unmanned vehicles considering multiple environmental factors and path planning, Scientific Reports 15:9957. Study area: Dengfeng City, China. What they did:

- Took 11 factors (elevation, slope, terrain position, landform, land cover, soil, geological hazards, rainfall, snowfall, wind, visibility) and gave each a 0-1 passability score from lookup tables.
- Combined the scores with a **weighted average**, with weights chosen by a method called AHP (experts compare factors two at a time).
- Noticed the average can hide a deal-breaker, so they added a fix: any cell that is NO-GO in any single factor is forced to NO-GO.
- Built maps for one wheeled vehicle (M151 jeep) and one tracked vehicle (M1 tank) at different times, then planned routes with an improved A*.

What they did not do, and what we add: they used the conditions *at the time*, not a forecast; they never checked whether their routes would actually have worked; they treated vehicles one at a time, ignoring how vehicles damage the ground for each other; and their weighted average needs a patch to avoid hiding NO-GO cells.

5. What the forecasting work already found

Before this poster, the project compared many forecasting models on the task 'predict RCI 1 to 7 days ahead', trained on 2015-2021, tuned on 2022-2023, and graded on 2024-2025 (724 forecast start dates never seen in training).

5.1 One trick that made everything work

Ground firmness changes slowly: today's value is almost exactly yesterday's. If you ask a model to predict tomorrow's *level*, it learns to copy today and adds nothing. Instead, every model predicts the **change** from today, and today's value is added back afterwards. With that, every learned model beats the lazy 'same as today' forecast.

5.2 Which models were most accurate

Model	Error 1 day ahead	Error 7 days ahead	Better than 'same as today' (7 days)
Ensemble (TCN + Transformer + GRU)	5.76	**19.95**	+24%
TCN	5.76	20.60	+21%
GRU	6.14	20.81	+21%
Ridge (a simple linear model)	6.37	21.96	+16%
ARIMAX-Fourier (classical statistics)	**5.76**	22.32	+15%
Persistence ('same as today')	6.79	26.21	0%

Errors are RMSE in cone-index units. Plain words: one day ahead, old-fashioned statistics is as good as anything; a week ahead, the neural networks pull clearly ahead.

5.3 But the most accurate model is not the best decision-maker

The ensemble has the smallest average error, yet it caught only **54%** of NO-GO days. Why? Average error rewards predicting values near the middle, and NO-GO days are the rare extreme ('the tail'). So 'most accurate' and 'safest' are different questions.

The fix is a **safety margin**: subtract a few units from the forecast before comparing it with the vehicle's VCI, so borderline days get called NO-GO. The margin is tuned per vehicle.

Decision: the model to use

Average of a GRU (128 units, 2 layers) and Ridge, minus a per-vehicle safety margin. For the 46-tonne tank class this caught about **90.6%** of NO-GO days with about 31 false alarms per year, and 84-94% of all three-way calls correct, depending on how far ahead. It has about 178,000 parameters and takes 0.68 MB of storage, small enough to run on a vehicle computer.

Why this pair works: the GRU learns patterns from the raw sequence of past days; Ridge is a straight-line model on hand-made features. They make *different* mistakes, so averaging them cancels some errors out. Adding a third neural network gained nothing. Making the GRU wider helped more than any change of architecture; attention layers and bidirectional variants were tried and made things worse.

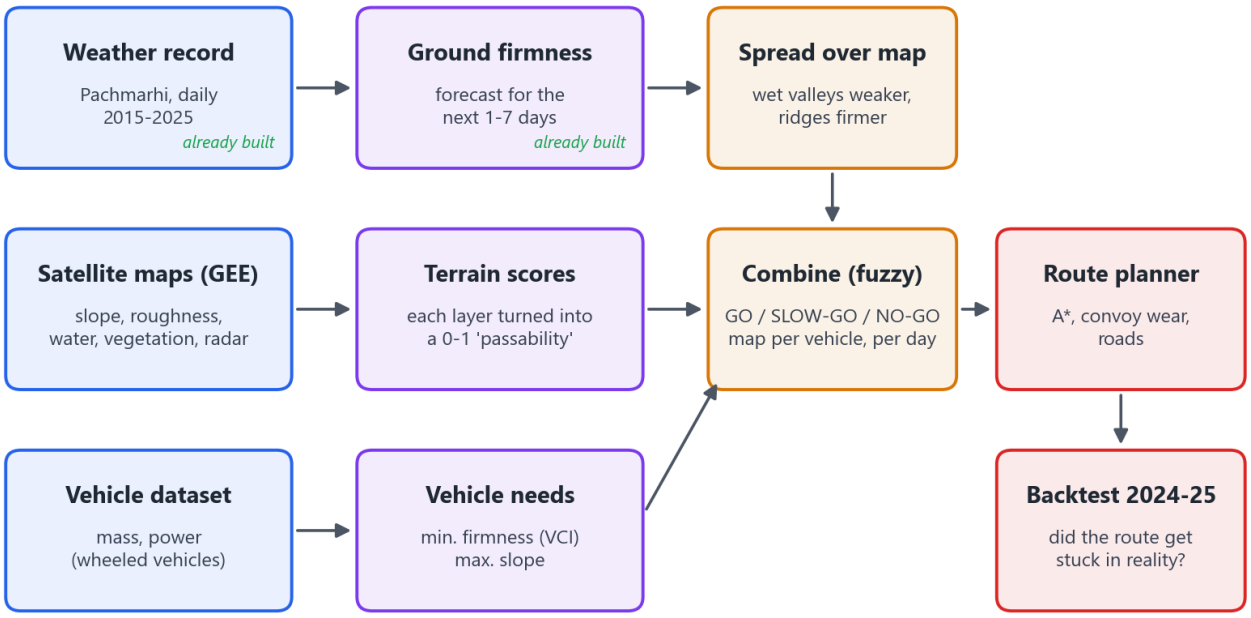
Correction to an earlier claim

An earlier summary (and the current README and RESULTS.md files) recommends the TFT model's 'P10' forecast, which caught 91.4% of NO-GO days. Follow-up tests in September rejected it. It reaches that number by calling NO-GO far too often (27.6% of days, against a true rate of 19.2%), so only 63.5% of its NO-GO calls are right. Its safety margin also changed a lot between years: tuned on 2024 it needed 21.0 units, tuned on 2025 only 10.5. **Do not quote the 91.4% figure.** The README and RESULTS.md need updating before the abstract is written.

5.4 How margins are checked

A margin tuned and tested on the same year will always look good. So each margin is tuned on 2024 and tested on 2025, then the other way round. A margin that is very different in the two directions cannot be trusted.

6. The plan, step by step



Blue = inputs we have. Purple = first processing. Amber = combining. Red = planning and testing.

Figure 3. The whole pipeline. The first two boxes in the top row exist already; everything else is built for the poster.

Step 1. Vehicle profiles

For each wheeled vehicle we need two numbers: **how firm the ground must be** (VCI1 and VCI50) and **the steepest slope it can climb**.

VCI1 comes from the US Army's standard empirical formula, the **Mobility Index** method, as written in test procedure TOP 02-2-619A. It turns a vehicle's weight, tyres, ground clearance, power and gearbox into one number. The dataset gives weight and power; tyre size and ground clearance are missing, so we assume typical values for each weight class (for example 4×4 with small tyres for light vehicles, 6×6 with large tyres for heavy trucks) and say so in the abstract. The slope limit comes from power-to-weight: underpowered, heavy trucks get a lower limit.

The exact formula (optional)

VCI1 = [11.48 + 0.2·MI − 39.2 / (MI + 3.74)] × DCF, for MI up to 115

MI = (CPF·WF / (TEF·GF) + WLF − CF) × EF × TF

CPF = axle load / (0.5 × tyres per axle × tyre diameter × tyre width) (contact pressure); WF = weight factor; TEF = (10 + tyre width) / 100; WLF = axle load / 2000; CF = clearance / 10; GF, EF, TF = 1.05 if tyre chains / under 10 hp per tonne / manual gearbox, else 1.0; DCF corrects for tyre deflection. Loads in lb, lengths in inches, VCI in psi. Source: Roșca et al. (2018).

Step 2. Terrain scores from the satellite layers

Each layer becomes a 0-1 score for each vehicle:

- Slope: 1 on flat ground, falling to 0 at the vehicle's slope limit (smoothly, see Section 7.5).
- Water (from NDWI or a land-cover map): 0. Wheeled vehicles do not swim.
- Dense forest (from NDVI or land cover): 0 for wheeled vehicles, following Liu et al.'s table.
- Roughness: lowers the score; it slows vehicles but rarely stops them.
- **Roads**, new: we add the road network from **OpenStreetMap**. Road cells score 1 whatever the soil, so the planner can choose between 'the long way on the road' and 'a shortcut across country'.

For a cleaner land-cover input we also ask GEE for **ESA WorldCover**, a free 10-metre global map with classes such as tree cover, cropland, built-up and water.

Step 3. Spreading the forecast across the map

The ground-firmness forecast is for a single 9 km weather cell, so on its own it would make the whole map wet or dry at once. In reality, low, flat valley floors collect water and stay soft longer than ridges. We adjust the single value cell by cell using:

- a **wetness index** computed from the elevation model (low, flat cells that collect water score as wetter), and
- **Sentinel-1 radar brightness**, which rises when the surface is wetter.

Wetter-looking cells get weaker ground, ridges firmer. This is a simple rule of thumb with no field measurements behind it, and the abstract will say so.

Step 4. Combine into GO / SLOW-GO / NO-GO maps

For every vehicle, every forecast day and every cell, the scores are combined into one passability value and one label. Instead of Liu's weighted average we use fuzzy-logic combinations (Section 7.5), which automatically make a cell NO-GO if any single factor is a deal-breaker. The result is a stack of maps: one per vehicle, per day ahead, per forecast date.

Step 5. Planning routes

A* finds the cheapest route from start to goal on these maps, three ways:

Planner	Map it uses	Role
Static	Terrain only, no weather	Naive baseline
Today	Today's ground conditions	What Liu et al. do
Forecast-aware (ours)	The forecast for the day the vehicle reaches each part of the route	Our method

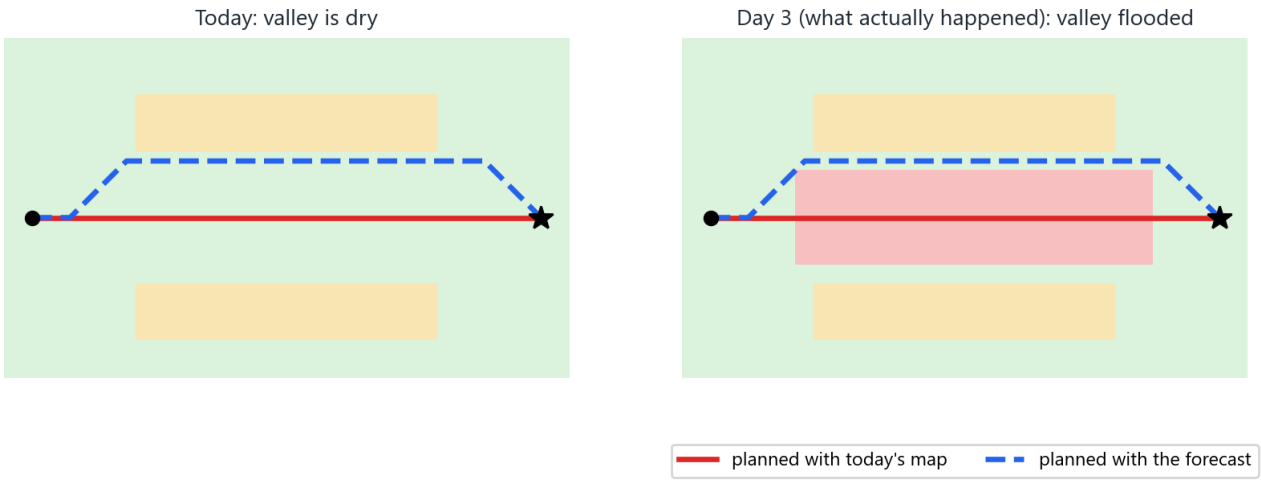


Figure 4. The idea in one picture (a made-up example, not a result). On the left, today's map shows a dry valley, so the 'today' planner (red) drives straight through it. The forecast knows rain is coming, so the forecast-aware planner (blue dashes) goes around. On the right, three days later, the valley really is NO-GO and the red route is stuck.

Step 6. Vehicles changing the terrain: the convoy experiment

Every vehicle that drives over soft ground churns it, so the next vehicle needs firmer ground than the first. This is exactly what VCI1 (one pass) versus VCI50 (fifty passes) captures. We let each cell count how many vehicles have crossed it, and raise the firmness it needs as the count grows.

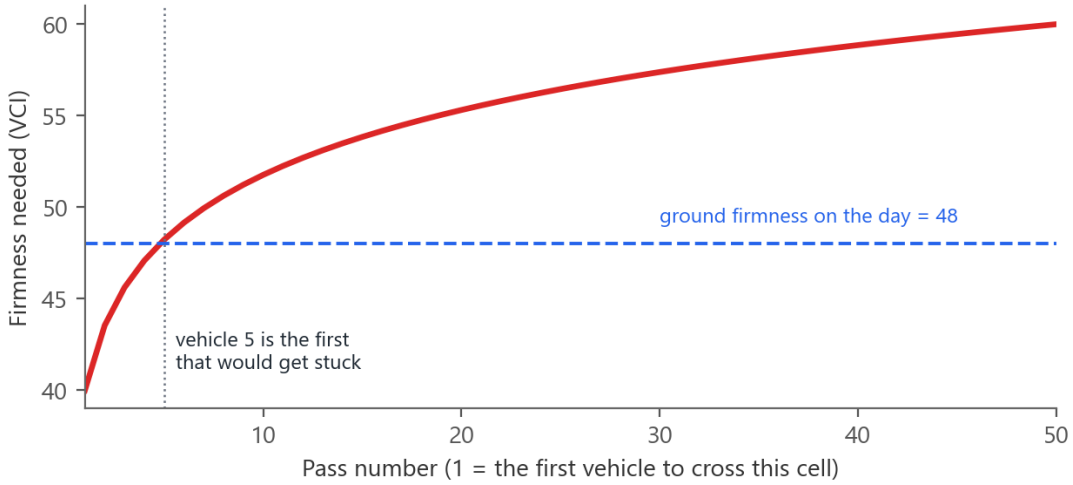


Figure 5. For a truck with VCI1 = 40 and VCI50 = 60, the firmness needed rises with each pass. On a day when the ground measures 48, the first four vehicles get through and the fifth gets stuck.

The exact formula (optional)

Firmness needed on pass n = VCI1 + (VCI50 − VCI1) × ln(n) / ln(50). This curve shape is our own assumption, chosen so that pass 1 gives VCI1 and pass 50 gives VCI50; it is not a published law.

We then compare two ways of moving a convoy of, say, ten vehicles:

- **Follow the leader**: everyone takes the first vehicle's route. Simple, but the track turns to mud partway down the column.

- **Wear-aware**: each vehicle plans its own route knowing where earlier vehicles have been, and shares its own track with the others. The convoy spreads across parallel lanes.

Several vehicles planning on one shared map that they change as they drive, while passing updates to each other over a network, is a distributed computing problem. That is the core of the ICDCN story.

Step 7. Proving it works: backtesting

We pick many start-and-goal pairs across the map and replay every forecast date in 2024-2025. Each planner makes its route, then we check that route against the conditions that *actually* happened on the travel days. Two scores, always reported together (the same logic as recall and precision in Section 2.5):

- **Route failure rate**: how often a route crossed a cell that had really become NO-GO, meaning the vehicle would have been stuck.
- **Detour cost**: how much longer the route was than necessary. A planner that never moves never fails, so this keeps it honest.

Results are split by vehicle weight and by season. The expected story is that heavy trucks at the start of the monsoon benefit most from planning with a forecast.

Step 8. Edge numbers

To support the 'runs on the vehicle' claim we measure: model size; time to make a forecast and re-plan on an ordinary laptop processor (no graphics card); and message size. A 7-day forecast is a few dozen numbers, so it can be sent over a weak radio link, whereas full maps cannot.

7. Your questions, answered

7.1 Should we use JEPA?

JEPA (Joint-Embedding Predictive Architecture) is a way to teach a neural network about images *without labels*. You hide part of an image and the network learns to predict a compact summary of the hidden part from the visible part. Doing this over millions of images teaches it general visual knowledge, which can then be reused for a labelled task.

> **Answer: not for this poster**
>
> JEPA needs (a) a large amount of unlabelled imagery and (b) a labelled task at the end to show that it helped. We have three seasonal snapshots of one area and no labels, so there is nothing to train it on properly and nothing to measure it against. In five days it would be a weakness. It works well as a **future-work** line on the poster: a JEPA-style model trained on years of Sentinel-1 and Sentinel-2 images could learn how the terrain changes over time.

7.2 Will the forecasting models improve if retrained with the new data?

Answer: no retraining needed

The forecaster predicts tomorrow's ground firmness at Pachmarhi from past weather. The new data does not add information about that: the maps are three snapshots that never change over time, and the vehicle file only changes the GO / NO-GO cut-offs. What **does** need redoing is the **safety margin for each new vehicle**, because margins are vehicle-specific (reusing the tank's margin for a light vehicle cost 10 accuracy points). That takes minutes and involves no retraining.

If one day you export a Sentinel-1 radar *time series* from GEE (a new image about every 12 days since 2015), that would be new information about soil wetness and worth testing as an extra input. Not for this deadline.

7.3 Should we train a separate model for GO / SLOW-GO / NO-GO?

Answer: no

Four reasons:

- The labels are just RCI with cut-offs applied. A classifier would learn a blurrier copy of what the forecaster already predicts.
- One forecaster serves every vehicle, because only the cut-offs change. Classifiers would need one model per vehicle: about 15 for the wheeled set.
- NO-GO days are rare, which makes classifiers harder to train well.
- For the map, there are no ground-truth labels at all (nobody drove the plateau and wrote down where they got stuck), so there is nothing to train a map classifier on.

The per-vehicle safety margin already does the decision job, and fuzzy logic (7.5) provides the soft in-between classes.

7.4 Can we model how vehicles change the terrain?

Level	What it means	Feasible in 5 days?
Pass-count wear	Each cell remembers how many vehicles crossed it; the firmness needed rises from VCI1 towards VCI50 (Step 6)	**Yes.** About a day of work, and our main new contribution
Ruts that last and recover	Ruts persist for days and slowly recover after rain	No. Needs rut-depth models we cannot check. Future work.
Full soil physics	Simulating how the soil compresses under each tyre (Bekker / Wong terramechanics)	No. Needs soil measurements we do not have.

7.5 Can we use fuzzy sets?

Yes, and it fits very well, because the 0-1 scores **already are fuzzy memberships**. It improves on Liu et al. in four ways.

(a) Smooth curves instead of step tables

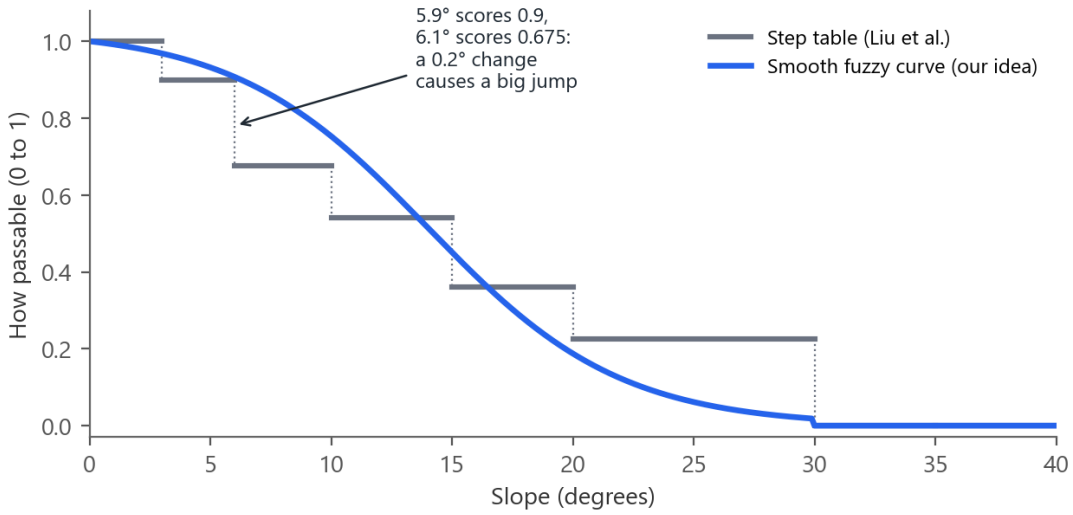


Figure 6. Liu et al.'s slope table for wheeled vehicles (grey) jumps at fixed boundaries, so tiny changes in slope can flip a decision. A smooth membership curve (blue) avoids that.

(b) A combination that cannot hide a deal-breaker

Take one cell: perfect slope, perfect soil, good weather, but it is a lake. Score each factor:

Combination method	How it works	Lake cell's result
Weighted average (Liu)	Add up the scores × weights	**0.81**, which looks fine!
Fuzzy AND, minimum	Take the weakest score	**0**, NO-GO, correct
Fuzzy AND, product	Multiply all scores	**0**, NO-GO, correct
OWA (ordered weighted average)	A dial that slides between 'minimum' and 'average'	Depends on the dial; we test where the best routes come from

Liu et al. noticed this weakness and added a separate fix afterwards. Fuzzy AND operators handle it automatically, as part of the method.

(c) SLOW-GO as a matter of degree

A cell can be 70% GO and 30% SLOW-GO, and simple readable rules can be written, such as 'IF soil is weak AND slope is steep THEN NO-GO'.

(d) How safe a whole route is

Multiply the scores of every cell along a route to get its overall reliability. A route through cells scoring 0.9, 0.9 and 0.5 has reliability 0.9 × 0.9 × 0.5 = 0.405. Setting each cell's cost to −log(score) turns that multiplication into addition, so ordinary A* directly finds the most reliable route. A NO-GO cell (score 0) gets infinite cost, so it is never used.

For the poster we compare all of these combination methods in the backtest. That comparison costs almost nothing extra to run.

7.6 Do the vehicles have to be Indian, or Class B?

Neither. This is preliminary, so we use the dataset's wheeled vehicles of any origin (Section 4.4). If 'vehicles that can stay on roads' meant something other than 'wheeled vehicles', that is the one thing to correct.

8. Limitations to be honest about

A poster is stronger when it states its weaknesses itself. These go in the abstract in brief.

Limitation	Why it matters	What we do
Ground firmness is computed, not measured	There are no cone-penetrometer readings from Pachmarhi, so all 'actual' conditions are model output	State it; field readings are future work
One 9 km weather cell	Real firmness varies over metres	Spread it with the wetness index and radar (Step 3), and say it is a heuristic
Vehicle data from a video game	Masses and powers are not certified figures	Clean it, disclose the source, assume tyre sizes by weight class
Doctrinal VCI thresholds	Not measured for these exact vehicles	Use the standard formula; state assumptions
Rainfall source changes in 2025	IMD gauges until 2024, ERA5 in 2025, with a different definition of 'a day'	Already documented; mention it
Convoy wear curve is our assumption	The ln(n) shape between VCI1 and VCI50 is not a published law	Say so; test whether conclusions change with a different shape
Three seasonal satellite snapshots only	Surface state within a season is treated as fixed	Fine for a poster; time series are future work
Units mislabelled as kPa	Would confuse a soil-mechanics reviewer	Relabel as cone-index units

9. Schedule and what is needed from you

Day	Work
Sat 10 Oct	Clean the wheeled-vehicle table and compute VCI1 / VCI50; write the GRU + Ridge + margin script; tune margins per vehicle (both-years check)
Sun 11 Oct	Load GeoTIFFs and OpenStreetMap roads; build fuzzy terrain layers; spread the forecast across the map
Mon 12 Oct	Three A* planners; compare combination methods
Tue 13 Oct	Convoy wear experiment; full backtest over 2024-2025
Wed 14 Oct	Figures, edge timing, write the 2 pages in the ACM template
Thu 15 Oct	Proofread and submit on EasyChair

Needed from you

- **GeoTIFF re-export from GEE**: the same layers, same area, all on one shared grid (30 m is fine), saved as numbers, not pictures. Please add **ESA WorldCover** (`ESA/WorldCover/v200`) and the **elevation model itself**. One file per season is ideal.
- **Confirm** that 'vehicles that can stay on roads' means wheeled vehicles.
- **Author names and affiliations** for page 1 (review is single-blind, so they are shown).
- **Who will present in person** at NITK Surathkal, 4-7 January 2027, and whether registration and the possible ACM article fee are covered.

A minimal GEE export looks like this (variable names will match your own script):

```
var stack = slope.addBands([tri, ndvi, ndwi, bsi, vv, vh])
  .addBands(ee.ImageCollection('ESA/WorldCover/v200').first())
  .addBands(dem).toFloat();
Export.image.toDrive({image: stack, description: 'pachmarhi_monsoon',
  region: aoi, scale: 30, crs: 'EPSG:32644', maxPixels: 1e10});
```

10. What the 2-page abstract will contain

Part	Content
Title	Something like: *Forecast-aware, wear-aware route planning for wheeled military vehicles at the edge*
Problem (¼ column)	Ground firmness changes daily; planners use today's map; forward units have no network link
Method (¾ column)	Edge forecaster (GRU + Ridge), fuzzy terrain combination, forecast-aware A*, shared pass-count wear for convoys
Figure 1	The pipeline as one horizontal strip
Figure 2	One vehicle's NO-GO map on a dry day and a monsoon day, with the three planners' routes
Table	Route failure rate and detour cost for the three planners, light vs heavy vehicles; follow-the-leader vs wear-aware convoy
Systems line	Model size, processor time, bytes per forecast broadcast
Limitations and future work	Field cone readings; Sentinel time series with a JEPA-style model; persistent ruts

A. Glossary

Term	Meaning
A*	An algorithm that finds the cheapest route on a grid quickly, by exploring the most promising cells first.
AHP	Analytic Hierarchy Process. A way of setting weights by having experts compare factors two at a time.
Backtest	Testing a method on past data as if it were being used live at the time, then checking against what really happened.
Cone index / RCI	A measure of ground firmness from pushing a cone into the soil. RCI also accounts for the soil weakening under traffic.
Edge computing	Running software on a device in the field instead of in a distant data centre.
ERA5 / ERA5-Land	Global reconstructions of past weather and land conditions, made by combining a computer model with observations.
Fuzzy set / membership	A group that things can belong to partly, with a degree between 0 and 1.
GEE	Google Earth Engine, a free platform for processing satellite data.
GeoTIFF	An image file that stores real measured values plus map coordinates.
GRU	Gated Recurrent Unit: a neural network that reads a sequence (here, past days) one step at a time.
Horizon (h)	How many days ahead a forecast looks.
JEPA	A way of pre-training image models without labels by predicting summaries of hidden image parts.
Mobility Index (MI)	A US Army formula that turns vehicle specifications into a single number used to estimate VCI.
NDVI / NDWI / BSI	Satellite indices for vegetation, water and bare soil.
OWA	Ordered Weighted Averaging: a way of combining scores with a dial between 'take the minimum' and 'take the average'.
Persistence	The forecast 'tomorrow equals today'. The baseline every model must beat.
Precision / recall	Of the warnings given, how many were right / of the real events, how many were warned about.
Ridge regression	A simple straight-line model that is protected against over-fitting.
RMSE	Root mean squared error: the typical size of a forecast's mistake.
Safety margin	An amount subtracted from the forecast so borderline days are treated as NO-GO.
Sentinel-1 / Sentinel-2	European satellites. Sentinel-1 is radar (sees through cloud); Sentinel-2 is optical (like a camera).
Skill	Percentage improvement of a model's error over persistence.
Slope / TRI	Steepness of the ground / how rough or bumpy it is.
Trafficability	Whether, and how easily, a particular vehicle can cross a particular piece of ground.
VCI1 / VCI50	The ground firmness a vehicle needs to pass once / to pass fifty times along the same track.

Term	Meaning
Wetness index	A number from the elevation map showing where water collects (low, flat places).

B. References

[1] ICDCN 2027 Call for Posters. cs3-lab.github.io/icdcn/submission/call-for-poster.html

[2] ICDCN 2027 Call for Papers, Accepted Papers and Workshops pages. cs3-lab.github.io/icdcn/

[3] ICDCN 2026 accepted posters (list kept by Osaka University Mobile Computing Lab). mc.net.ist.osaka-u.ac.jp

[4] ICDCN 2025 Call for Posters and Demos. cse.iith.ac.in/icdcn-2025/call_for_poster.html

[5] Liu, Q., You, X., Zhang, X., Zuo, J. (2025). Construction of dynamic trafficability map for unmanned vehicles considering multiple environmental factors and path planning. *Scientific Reports* 15:9957. doi:10.1038/s41598-025-94377-4

[6] Roșca, P., Ciobotaru, T., Olaru, G., Puică, C. (2018). Considerations regarding Vehicle Cone Index for fine-grained soil. *Journal of Military Technology* 1(2). doi:10.32754/JMT.2018.2.07

[7] Wong, J. Y. *Theory of Ground Vehicles*. Wiley. (Background on terramechanics and the Mobility Index.)

[8] ESA WorldCover 10 m land cover, available in Google Earth Engine as ESA/WorldCover/v200.

[9] Soft-skinned ('B') vehicle definition: en.wikipedia.org/wiki/Soft-skinned_vehicle

[10] Project's own files: README.md, metadata_sheet.csv, outputs/RESULTS.md, traffic/data.py.